UAM SingPass or CorpPass Stateless SAML Config and API Integration de - 6.0.1-GA (AI)

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1. Preface - UAM SingPass or CorpPass Stateless SAML Config and API Integration Guide

The AccessMatrix Universal Authentication Management (UAM) system supports integration with SingPass/CorpPass using the SAML 2.0 protocol. This guide provides information on how to do so.

Intended Reader

This guide is intended as a intended as a reference for security administrators, i-Sprint's Professional Service personnel, and developers.

Document Scope

This document provides instructions on how to integrate SingPass/CorpPass using the SAML 2.0 protocol with the AccessMatrix Universal Access Management (UAM) system. It also guides you on how to perform the application integration via the UAM API.

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix Product Suite:

- *UAM Administration Guide*
- *UAM WSA Reverse Proxy Implementation Guide*
- *UAM SAML Implementation Guide*
- *AM5 XML-RPC and SOAP API Reference*

2. Introduction to UAM SingPass or CorpPass Stateless SAML Config and API Integration

Due to the complexity of having a SingPass/CorpPass environment, this document shows you how to simulate a SingPass/CorpPass Identity Provider using a separate AccessMatrix UAM system for testing purposes. The sequence and architecture diagrams provided in this document are based on the simulated Identity Provider (IdP) environment.

i	**Note:** In this guide, the term AM5 refers to the AccessMatrix UAM system.

- **Identity Provider (IdP)**
  SingPass/CorpPass is the identity provider which gives assurances of the identity of a user to the other party. The identity provider is responsible for:
  - Managing users and their identities.
  - Issuing credentials.
  - Handling user administration.
  - Authenticating the user.
  - Vouching for the user's identity with the service provider.
- **Service Provider (SP)**
  Agency application is the service provider (sometimes called the relying party or resource partner or consumer) and the "validating party" in a transaction. The service provider is responsible for:
  - Controlling access to services.
  - Validating the asserted identity information from the identity provider (typically by way of verifying a digital signature).
  - Providing access based on the asserted identity.
  - Managing only the locally relevant user attributes but not an entire user profile.

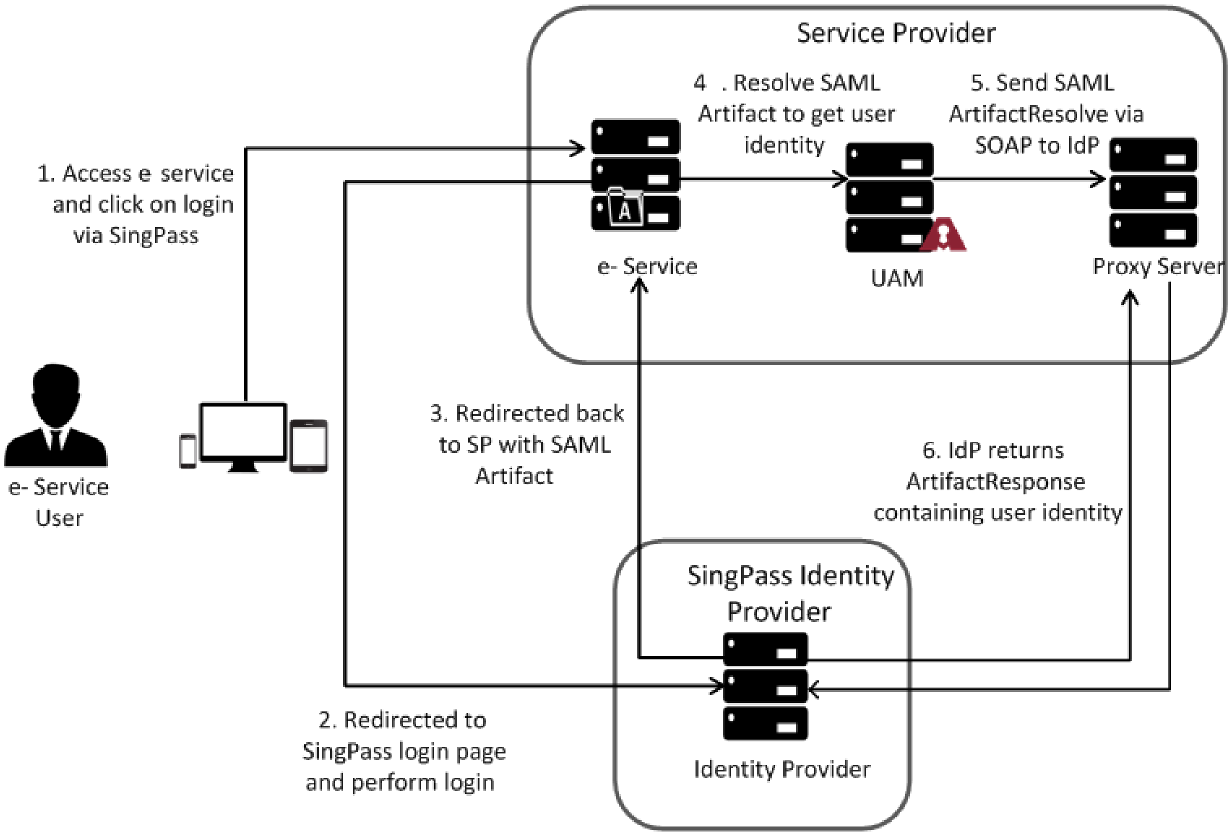
SAML Assertions, Protocols, Bindings and Profiles

- **Assertions** - contains the actual user data including the user's attributes (e.g. NRIC) and entitlements (e.g. roles, UEN, etc.). Assertion details are pre-agreed between the SingPass/CorpPass and agencies.
- **Protocols** - defines the type of request that is being sent to the other party in the transaction, i.e. when a service provider wants to retrieve the user information from Identity Provider (IdP)

after receiving the artifact, it sends an <ArtifactResolve> request to IdP as defined in the Artifact Resolution protocol.

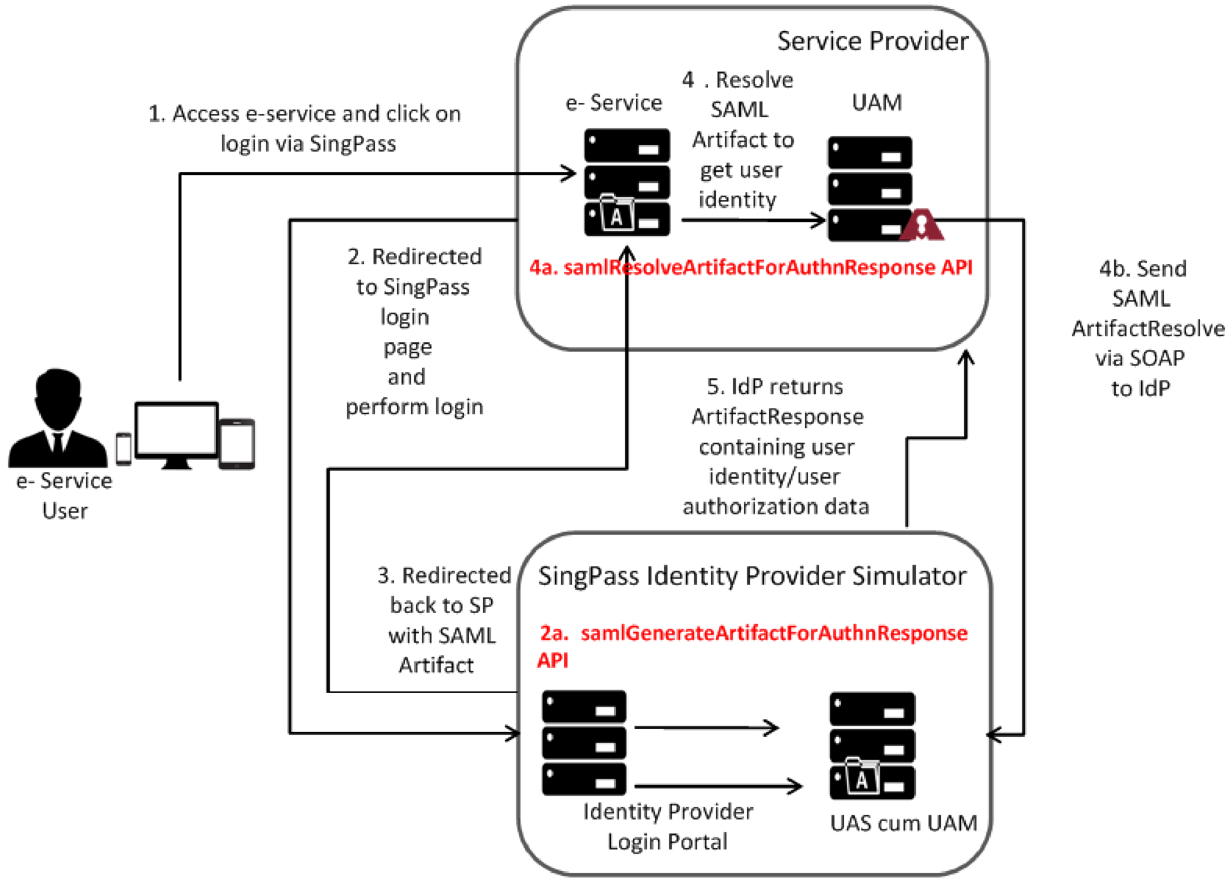
- **Bindings** - refers to the communication protocols used in exchanging of requests and assertions. SingPass/CorpPass uses **HTTP POST** and **SAML SOAP** bindings.
- **Profiles** - refers to a combination of bindings, protocols and assertions that together defines how the federated identity data is exchanged. SingPass/CorpPass supports the Artifact Resolution profile.
- **Authentication Context** - provides the details on how a user has been authenticated with Identity Provider. This allows the agencies to identify if the user has used 1-factor or 2-factors during authentication.
- **Metadata** - refers to the data that is exchanged between Service Provider (SP) and Identity Provider (IdP) while establishing the trust relationship. This includes the settings as well as the crypto material.

Identity Provider Initiated Artifact Binding Flow



Identity Provider Initiated Artifact Binding Flow (with UAM Simulated Identity Provider)

The following figure shows the integration of Identity Provider initiated Artifact Binding (with UAM Simulated Identity Provider) using UAM API. The text in red specifies the API calls needed for the integration.



3. UAM SingPass or CorpPass Stateless SAML Settings

Setup Configuration

To integrate SingPass/CorpPass with UAM system, carry out the following steps:

1. Install UAM as Identity Provider (optional).

   Note: This is used only for demonstration or testing purpose to simulate an identity provider environment.

2. Install UAM as Service Provider.
3. Exchange metadata to establish trust connection with Identity Provider.
4. Configure the Service Provider and Identity Provider on both Identity Provider UAM and Service Provider UAM.
5. Fine tune the cache size for response and artifact in SAML Configuration of AccessMatrix UAM system.

Prerequisite

The following requirements are needed for Service Provider (SP) configuration and integration with Identity Provider (IdP):

- Identity Provider URL / Provider ID / Point of Contact Server
- Federation Name / SP Identity (Id)
- Identity Provider Metadata File
- SSL Server Authentication Certificate
- SSL Client Authentication Certificate (not required for Internet OOB)
- Signing Algorithm
- Encryption Algorithm
- SAML Assertion Settings
- Identity Mapping
- Session Timeout (seconds)
- Logout Request Lifetime (seconds)

Identity Provider (Simulated IdP)

Cryptographic Material:

- SSL Service Provider Client Certificate or Certificate Authority (CA) of Service Provider Client Certificate for Artifact Resolution Service (public key)
- SSL Identity Provider Server Certificate for Artifact Resolution Service (private key)
- Service Provider Digital Signature Verification Key
- Identity Provider Signing Key
- Service Provider Certificate for Artifact Response Assertion Encryption

Service Provider

Cryptographic Material:

- SSL Service Provider Client Certificate for Artifact Resolution Service (private key)
- SSL Identity Provider Server Certificate or CA of Identity Provider Server Certificate for Artifact Resolution Service (public key)
- Service Provider Signing key
- Identity Provider Digital Certificate for Verification of Artifact Response
- Service Provider Key for Artifact Response Decryption

Configure AccessMatrix UAM System as Service Provider

> **Note:** Refer to Service Provider Configuration for the list of Service Provider attributes and to Identity Provider Configuration for the list of Identity Provider attributes.

Required settings during the setup:

Identity Provider (IdP) Configuration

- Identity
- Name
- URL
- Public Key to verify the signature of artifact response
- Signing Algorithm

- Artifact Resolution Service URL
- HTTP Proxy used for Artifact Resolution

Service Provider (SP) Configuration

- Identity
- Name
- URL
- SSL Client Certificate for Artifact Resolution Service
- Private Key for signing artifact resolve
- Signing Algorithm
- Private Key to decrypt assertion of artifact response
- Indicate assertion as the scope of response signature
- Assertion Consumer Service to consume SAML artifact

Others

- Place the SSL Server Authentication Certificate in the trust store of the application server.

> **Note:** For tomcat, configure this in the **setenv.bat** file within *<AM_ROOT>\tomcat\bin*. For example:
>
> Code
>
> ```
> "-Djavax.net.ssl.trustStore=C:/testtrust.keystore"  "-
> Djavax.net.ssl.trustStorePassword=yourtruststorepasswordpassword"
> ```

Advance Service Provider (SP) Configuration

- Configure the time window for assertion validity (based on assertion issue time).

Configure AccessMatrix UAM System as Identity Provider (simulated IdP)

> **Note:** Refer to Service Provider Configuration for the list of Service Provider attributes and to Identity Provider Configuration for the list of Identity Provider attributes.

Required settings during the setup:

Identity Provider (IdP) Configuration

- Identity

- Name
- URL
- Signing Key for Artifact Response
- Signing Algorithm
- Authentication Session Life Time of Assertion

Service Provider (SP) Configuration

- Identity
- Name
- URL
- Verify Digital Signature of Artifact Resolve with Service Provider's Public Key
- Encrypt assertion of artifact response with Service Provider's public key
- Encryption algorithm for the artifact response
- Subject\NameId Value of the artifact response
- Assertion as the response signature scope
- Indicate response encryption is required

Edit SAML Configuration (Optional)

To configure:

1. Navigate to **Administration Dashboard > Configuration > Unified SSO > Federated SSO > SAML > SAML Configuration**.
2. Click the **Edit** button if there is a need to change the default values.

Attribute	Description
Default Identity Provider	To denote the default Identity Provider referenced by XML-RPC API call where Identity Provider (IdPId) is not specified. Default: DefaultIdentityProvider
Cache	
Response life time	Life time of used response. This cache is to prevent reuse of response when AccessMatrix UAM system acts as a service provider (SP). Default: 900

Attribute	Description
Response Cache Size	Number of user response to be kept in cache. This cache is to prevent reuse of response when AccessMatrix UAM system acts as a service provider (SP). Default: 5000
Artifact life time	Life time of artifact. The cache is used when the AccessMatrix UAM system, acts as an Identity Provider to cache user identity together with the artifact. Default: 300
Artifact Cache Size	Number of artifacts to be kept in cache. The cache is used when AccessMatrix UAM system acts as an Identity Provider to cache user identity together with the artifact. Default: 5000

4. UAM SingPass or CorPass Stateless SAML Configuration

Identity Provider Configuration

To configure:

1. Navigate to **Administration Dashboard > Configuration > Unified SSO > Federated SSO > SAML > Identity Provider** and click **Create**.
2. Specify the necessary values for the following Identity Provider attributes:

<table>
<tr><th>Attribute Name</th><th>Description</th><th>Remark</th></tr>
<tr><td>*Identity Provider Id</td><td>Unique Id of the Identity Provider.</td><td></td></tr>
<tr><td>*Identity Provider Name</td><td>Name of the Identity Provider.</td><td></td></tr>
<tr><td>Description</td><td>Brief description for the Identity Provider.</td><td></td></tr>
<tr><td>Version</td><td>The Identity Provider revision. (uneditable)</td><td></td></tr>
<tr><td>*Id used as issuer</td><td>An Id to represent the entity IdP which is normally in URL form.</td><td></td></tr>
<tr><td colspan="3">General Tab</td></tr>
<tr><td colspan="3">Digital Signature</td></tr>
<tr><td>Key store</td><td>Key repository of the signing certificate.</td><td rowspan="3"><u>When AccessMatrix UAM system is Identity Provider:</u>
• The Private key in the key store is used for signing authentication response or assertion.
• The Public key is required to verify signature, and it is to be placed into a key store.
<u>When AccessMatrix UAM system is Service Provider:</u>
• The Public key for verifying the certificate should be in the key store of the AccessMatrix UAM System Identity Provider configuration.
Alternatively, an X.509 certificate (public key) can be configured with .pem extension (a Base64 encoded DER certificate) enclosed between -BEGIN CERTIFICATE- and -END CERTIFICATE- for signature verification. Refer to IdP public key string for additional details.</td></tr>
<tr><td>Key store passphrase</td><td>Passphrase of the key store for the signing.</td></tr>
<tr><td>Key alias</td><td>Alias of the key store used for signing.</td></tr>
</table>

Attribute Name	Description	Remark
		By default, it is "Response" that the SAML scope is using for signing the response. AccessMatrix UAM system also supports signing "Assertion". Refer to **Response Signature Scope** in Service Provider Configuration.
Signature Algorithm	Refers to the signing algorithm of the XML signature. Default: RSA_SHA1	
IdP public key string	When AccessMatrix UAM system is the Service Provider, this value can be configured as the Response's signature verification key to verify the SAML response's signature (signed by IdP).	
Identity Provider System Properties Tab		
Service URLs		
Artifact Resolve Service URLs	The Identity Provider service URL to receive Artifact Resolve request from Service Provider. **Note:** This configuration is needed when the AccessMatrix UAM system serves as the Service Provider.	Default value for Idp simulated environment using AccessMatrix UAM system: e.g. https://testserver:8443/am5/wpe/IDPArtifactResolution
Attribute Query Service URLs	The Identity Provider service URL to receive an Attribute Query request from Service Provider. **Note:** This configuration is needed when AccessMatrix UAM system serves as the Service Provider.	
Single Sign-On Service URLs	The Identity Provider service URL to receive an	

Attribute Name	Description	Remark
	Attribute Query request from Service Provider. **Note:** This configuration is needed when AccessMatrix UAM system serves as the Service Provider.	
Authentication Request		
Acceptance time (in seconds)	The difference between the time, the authentication request is issued and the time, the authentication request is received must be within this configured time. **Note:** Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.	
Authentication Request Required Signing	Set this to "true" if SAML request requires signing. **Note:** Referred by Service Provider when AccessMatrix UAM system serves as Service Provider. Default: false	
Authentication Response		
Assertion validity time (in seconds)	Used to compute Assertion Condition Not On Or After datetime (Response Issue time plus this validity duration). **Note:** Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider. Default: 60	The validity of SAML Response and SAML Response's Assertion is based on a time-based condition which specifies the validity time range. When the AccessMatrix UAM system is an Identity Provider and upon generating SAML Response, this attribute value would be used to derive the notOnOrAfter time condition of the Assertion Subject and the Response Subject. Time is always in UTC timezone. The time unit of this attribute is in seconds.

Attribute Name	Description	Remark
Assertion Start Time Validity Adjustment (for IdP initiated response only)	Used to compute Assertion Condition Not Before datetime (Response Issue time minus this validity duration) when the response is IdP initiated. **Note:** Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.	For IdP-initiated Response, when SAML Response is generated, its Assertion Subject's Not Before time is generated based on the issue time of the Response. Sometimes the response time on the servers for the service provider and the identity provider may not synchronize and the service provider fails to use the SAML Response. To resolve this issue, this configuration is introduced to allow buffer time to be included (Assertion Subject's Not Before time = issue time - Assertion Start Time Validity Adjustment (for IdP initiated response only)) to Assertion Subject's Not Before time. Time is always in UTC timezone. The time unit of this attribute is in seconds.
Advance Tab		
HTTP Proxy		
Enabled	To enable the use of HTTP Proxy for artifact resolution.	
URL	URL of the proxy server.	
Port	Port of the proxy server.	
User Name	User name to authenticate the proxy server.	
User Password	User's password to authenticate the proxy server.	
NTLM Authentication	This is to indicate whether NTLM authentication is used. If NTLM Authentication is enabled, Domain name must be specified. In addition to that, it may be needed to configure each AccessMatrix	

Attribute Name	Description	Remark
	server host domain name or IP address in **amsystem.properties** with the property key "am.saml.ntlm.host" for each instance.	
Domain Name	Domain name of AD used in NTLM authentication.	

Service Provider Configuration

To configure:

1. Navigate to **Administration Dashboard** > **Configuration** > **Unified SSO** > **Federated SSO** > **SAML** > **Service Provider** and click **Create**.
2. Specify the necessary values for the following Service Provider attributes:

Attribute Name	Description	Additional Remark
*Service Provider Id	Id of the Service Provider.	It can be the name of the federation with the following naming format: "SPCP_<Agency ID>_<E-SRVC ID>_saml20fed". **Note:** SPCP means SingPass/CorpPass.
*Service Provider Name	Name of the federation setup in Service Provider.	The following naming format is recommended: "SPCP_<Agency ID>_<E-SRVC ID>_saml20fed". **Note:** SPCP means SingPass/CorpPass.
Description	Brief description for the Service Provider.	
Version	The Service Provider revision. (uneditable)	
*Id used as issuer	An Id to represent the entity IdP which is normally in URL form.	
Web Browser SSO Tab		
Assertion		

<table>
<tr><th>Attribute Name</th><th>Description</th><th>Additional Remark</th></tr>
<tr><td>*Assertion Consumer Service URLs</td><td>Service Provider's URL to consume SAML assertion. This configuration is needed when AccessMatrix UAM system is IdP, to specify the Subject recipient in the SAML Response.</td><td></td></tr>
<tr><td>Assertion Acceptance time (in seconds)</td><td>The difference between the time the assertion is issued and the time the assertion is received must be within this configured time. This is used when AM is Service Provider, to indicate the tolerance time of response's assertion. It will validate response's assertion time where the difference of assertion issued time and response's assertion received time must be within the specified tolerance time.
Default: 240</td><td>Note: Referred by Service Provider when AccessMatrix UAM system serves as Service Provider.</td></tr>
<tr><td colspan="3">Name Id</td></tr>
<tr><td>Format</td><td>A constant to describe the NameID value.
Note: The important thing is to agree on the NameID format between IdP and Service Provider.
Most common: "Unspecified", "Email".
Default: Unspecified</td><td rowspan="3">NameID Value in the Subject of SAML Response is used to represent the identity of the user. In AccessMatrix UAM system's Service ProviderConfiguration, NameID Format defines the format of the user identity value and NameID Value defines the retrieval of the user identity value.</td></tr>
<tr><td>Value</td><td>Value to be retrieved from user object as user identity. If the user's identity is to be retrieved from User Response (e.g. user's UUID), configure this attribute to "User Response" instead.
Note: Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.</td></tr>
<tr><td>User Response to return as NameId value</td><td>Value to be retrieved from User Response as user identity. This setting is used when NameID Value is set as "User Response". User Response must be selected for the Value configuration.
Note: Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.</td></tr>
<tr><td colspan="3">Attributes</td></tr>
</table>

Attribute Name	Description	Additional Remark
Attributes to return in the assertion	Refers to the user information that is set as attributes as part of Response's Assertion. **Note:** Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.	SAML Assertion allows add-on payload to include user attributes or other information from IdP to Service Provider.
Access Control		
Allow All Users to Access	Enabling this feature allows all authenticated users to access this service provider. To control user access to the web applications or web services of a service provider, this setting must be disabled. To control user access, go to the **SAML Application Privilege** tab of the user's **View User** page to allocate. Default: true	**Note:** Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.
Others		
URL to initiate SSO	A URL that initiates SAML SSO operation. It is a URL to a servlet(s) for the SSO service or the SAML AuthnRequest Service.	**Note:** It is required only by IdP portal or USO Agent.
Is WSA Being Used?	Indicate if WSA is used or not. Default: false	**Note:** Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.
General Tab		
Response		
Response signature scope	Defines the scope of the SAML response to be signed, e.g. "Response", "Assertion" and etc. Default: Response	**Note:** Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.
Response encryption	Indicate whether to apply encryption to SAML response's assertion.	**Note:** Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.
Assertion Attribute		

<table>
<tr><th>Attribute Name</th><th>Description</th><th>Additional Remark</th></tr>
<tr><td>Assertion Attribute Unique Name</td><td>Name of SAML Assertion Attribute (e.g. "Name" or "FriendlyName") to use, to uniquely identify the SAML attribute.
Default: Name</td><td></td></tr>
<tr><td colspan="3">Digital Signature</td></tr>
<tr><td>Keystore</td><td>Key repository.</td><td rowspan="3">Keypair of XML signature for SAML Request when AccessMatrix UAM system is a service provider. The private key in the key store is used for SAML request signing. However, SAML request signing can only be enabled if the Authentication Request Required Signing attribute value of Identity Provider is set to "true". By default, the signature algorithm to be used is "RSA_SHA1".

AccessMatrix UAM system also supports "RSA_SHA256".</td></tr>
<tr><td>Keystore Passphrase</td><td>Passphrase of a key store for the signing.</td></tr>
<tr><td>Key alias</td><td>Alias of the key store used for signing.</td></tr>
<tr><td>Signature Algorithm</td><td>To change the algorithm of SAML request.
Default: RSA_SHA1.</td><td></td></tr>
<tr><td colspan="3">XML Encryption</td></tr>
<tr><td>Keystore</td><td>Key repository.</td><td rowspan="3">To configure keypair used for XML encryption.

<u>AccessMatrix UAM system as IdP:</u>
Public key used by IdP to encrypt SAML response's assertion.
Import public key of service provider to the key store of XML encryption, and configure the Response encryption attribute to "true".
<u>AccessMatrix UAM system as Service Provider:</u>
The private key used by a service provider to decrypt SAML response's assertion.</td></tr>
<tr><td>Keystore Passphrase</td><td>Passphrase of a key store for the signing.</td></tr>
<tr><td>Key alias</td><td>Alias of the key store used for signing.</td></tr>
</table>

Attribute Name	Description	Additional Remark
		• Import private key of a service provider to the key store to do XML encryption. Optionally, configure the **Service Provider public key string** instead of the key store. By default, the signature algorithm to be used is "RSA_SHA1". AccessMatrix UAM system also supports "RSA_SHA256". **Note:** Referred by Service Provider when AccessMatrix UAM system serves as Service Provider
Encryption Algorithm	To change the algorithm of SAML Response's assertion encryption. Default: AES_128	
SSL Client Authentication		
Keystore	Key repository.	SSL Client Authentication for SAML SOAP Binding. It could be used for Artifact Resolution Protocol and Attribute Query Protocol. This setting is to configure the client certificate which is to be used in the SSL Client Authentication. This configuration is needed, when: AccessMatrix UAM system is a service provider, Artifact Binding is used for Web Browser SSO, SOAP communication is required, and mutual authentication is required.
Keystore passphrase	Passphrase of a key store for the signing.	
Key alias	Alias of the key store used for signing.	
Others		
Service Provider public key string	When AccessMatrix UAM system is an Identity Provider, this value can be configured as: • "Request's signature verification key" - To verify SAML request's signature (signed by Service Provider).	**Note:** Referred by Identity Provider when AccessMatrix UAM system serves as Identity Provider.

Attribute Name	Description	Additional Remark
	• "Response encryption key" - To encrypt SAML response upon generating the corresponding response.	

SOAP Client Connection Pool Configuration

To configure, add the following attributes to **amsystem.properties**

(*<AM_ROOT>/tomcat/webapps/am5/WEB-INF/classes/*):

Properties (.properties files)

```
soap.proxy-port=-1
soap.preemptive-authentication=true
soap.content-char-set=UTF-8
soap.connection-time-out=5000
soap.send-buffer-size=4096
soap.receive-buffer-size=16384
soap.tcp-no-delay=true
soap.max-connections-per-host=5
soap.max-total-connections=20
soap.connection-retry-attemps=0
```

SOAP HTTP Client Connection Configuration

To configure, add the following attributes to **amsystem.properties**

(*<AM_ROOT>/tomcat/webapps/am5/WEB-INF/classes/*).

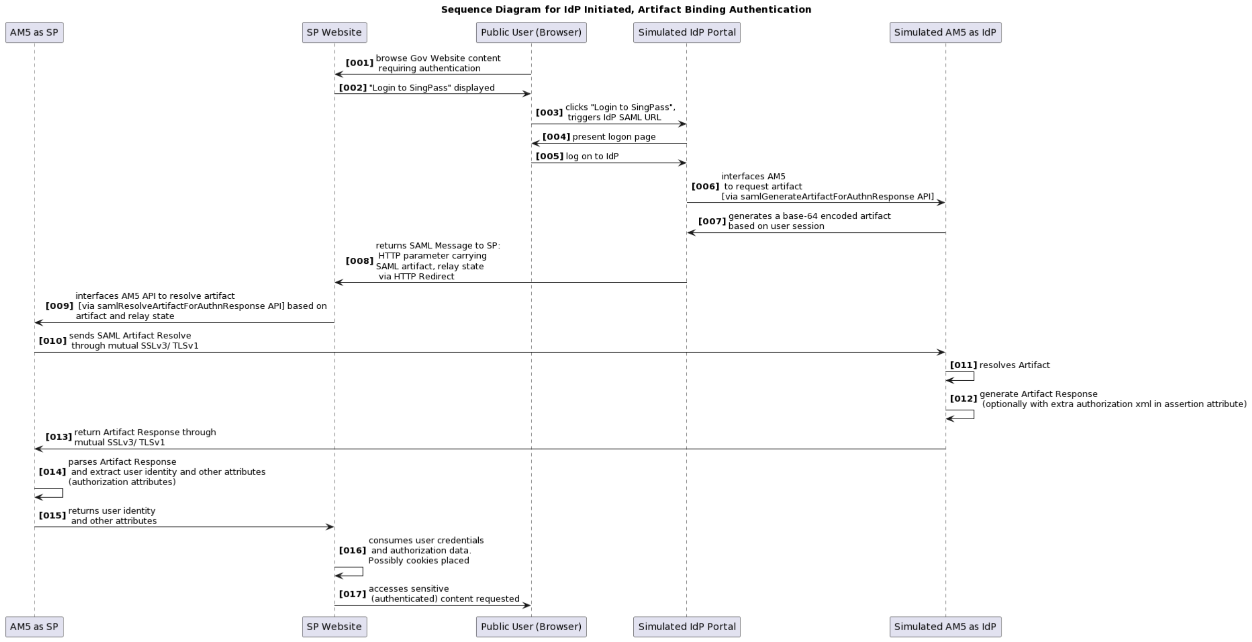
Properties (.properties files)

```
soap.enabled-protocols=TLSv1.1
soap.enabled-cipher-
suites=TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA,
TLS_RSA_WITH_AES_128_CBC_SHA
#default value is true
soap.enabled-persistence-connection=false
```

5. UAM SingPass or CorPass Stateless SAML APIs

This section contains the API use cases and its sample codes for service provider and identity provider integration.

Sequence Diagram (AccessMatrix UAM System Identity Provider as Simulator)



i	**Note:** AM5 in the diagram refers to the AccessMatrix UAM system, SP refers to Service Provider and IdP refers to Identity Provider.

- **Step 006**

  Interface with am5 (as IdP) to generate artifact.

- **Step 009**

  Interface with am5 (as SP) to resolve artifact and extract user identity and other attributes (authorization attributes).

- **Step 012**

  For CorpPass third party user use case or CorpPass user use case, extra attribute values like authorization XML of user may be required. To customize, refer to Authorization XML for CorpPass User for details.

XML-RPC Sample Codes

AccessMatrix UAM System as Identity Provider

- **XMLRPC API**

  AuthzService.samlGenerateArtifactForAuthnResponse
- **Customization API for simulation**

  Set authorization data (e.g. CorpPass User Authorization XML or CorpPass Third Party Authorization XML) for assertion attribute of user artifact response.

<u>**Generate Artifact For Authenticated User**</u>

Synopsis:

Generate artifact for Authentication Response based on user's login session token.

		XML-RPC	**SOAP**
Returns		String	Not Available
Since	AM ver		
	Client ver		
Input	**Desc**	**Type/ Field name if different**	
sessionToken	session of the authenticated user	String	
SPIssuer	Service Provider Issuer	String	
params	Additional parameters	struct	

Sample Code:

XML-RPC (Java)

Java

```
        ApiProxy proxy = null;
        Map<String, String> params = new HashMap<String, String>();
         try {
```

XML-RPC (Java)
``   proxy = ApiProxyFactory.createProxy("SimpleFactory");   String artifact = proxy.getAuthzService().samlGenerateArtifactForAuthnResponse(sessionToken, SPIssuer, params);   } catch (XmlRpcException e) {   fail(e.code + "," +e.getMessage());   } finally {   proxy.close();   } ```

Authorization XML for CorpPass User

Synopsis:

Implement this interface (and make it a plugin) to set authorization data (e.g. CorpPass User Authorization XML or CorpPass Third Party Authorization XML) for assertion attribute of user artifact response. Set the attribute name and value of the SAML assertion to the key value "Map<String, Object>".

Class Name:

Java

```
com.isprint.am.core.saml.ISAMLResponseAttributeInterceptor
```

Method:

Java

```
public Map<String, Object> intercept(IUser
responseSubjectUser);
```

AccessMatrix UAM System as Service Provider

- **XMLRPC API**

  AuthnService.samlResolveArtifactForAuthnResponse

Resolve Artifact With API

Synopsis:

Resolve artifact with Identity Provider via backend channel to get SOAP Artifact Response, then further parsing and validating the response to have it returned as AuthnResponseTO.

Returns:

- Content of SAML response in the form of AuthnResponseTO.
- On error, XML-RPC fault will be returned.

		XML-RPC	SOAP
Returns		AuthnResponseTO	Not Available
Since	AM ver		
	Client ver		
Input	**Desc**	**Type/ Field name if different**	
sessionToken	session; agent session Id	String	
spId	Service Provider Id used in AM5	string	
idpId	Identity Provider Id used in AM5	string	
artifact	artifact from Identity Provider to be resolved	string	
relayState	target URL of Service Provider	string	
params	Additional parameters	struct	

Sample Code:

XML-RPC (Java)

Java

```java
try {

        factory = ApiProxyFactory.registerFactory(
                "SimpleFactory",
            new String[]{url},
            10000);
    proxy = factory.createProxy();
        AuthnResponseTO rst  =
proxy.getAuthzService().samlResolveArtifactForAuthnResponse("","SPId",
"IdpId", artifact, "",new HashMap());
        System.out.println("User Identity : " +
```

XML-RPC (Java)
`` rst.getUserIdentity()); } catch(XmlRpcException e){   e.printStackTrace(); } catch (MalformedURLException e) {   // TODO Auto-generated catch block   e.printStackTrace(); } ```